

Exercise 2: SQL FOUNDATIONSSQL Aggregates Functions and Operators

students_table

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

01. List all distinct departments in the students table.

EXPECTED COLUMNS: department

SELECT DISTINCT department
FROM students_table; OUTPUT

results_table

department
IT
HR
Finance

02. Get the average age of students per department

EXPECTED COLUMNS: department, student_count

SELECT department, avg-age
AVG(age) AS student_count
FROM students_table
GROUP BY department;

avg-age student_count	department
20.5	IT
22	HR
23	Finance

Q3. Show departments with more than 1 student
EXPECTED COLUMNS: department, student_count.

```
SELECT department,  
COUNT(student_id) AS student_count  
FROM students_table  
GROUP BY department  
HAVING COUNT(student_id) > 1;
```

results-table

student_count	department
2	IT
2	HR

Q4. Get all students whose age is between 21 and 23
EXPECTED COLUMNS: student_id, name, age, department.

```
SELECT student_id,  
name,  
age,  
department  
FROM students_table  
WHERE age BETWEEN 21 and 23;
```

results-table

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05. List all students in the IT or HR department who are older than 21.

EXPECTED COLUMNS: student_id, name, age, department

SELECT *

FROM students_table

WHERE department = 'IT' OR department = 'HR'

AND age > 21

Or

SELECT *

FROM Students table

WHERE department IN ('IT', 'HR') AND age > 21;

results-table

student_id	name	age	Department
2	Bob	22	HR
5	Eve	22	HR

courses-table

Course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

06. Show all credits per department, only for departments with more than 5 total credits

SELECT department,
SUM(credits) AS total_credits

FROM courses_table

GROUP BY department

HAVING SUM(credits) > 5;

results_table

department	total_credits
IT	11

07. List all courses that do not have 4 credits.

EXPECTED COLUMNS: course_id, course_name, department, credits

SELECT *

FROM courses_table

WHERE credits NOT = 4;

or

SELECT *

FROM courses_table

WHERE credits <> 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08. Show the top 3 courses by credits in descending order.

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Q8. Show the top 3 courses by credits in descending order.
EXPECTED COLUMNS: course_id, course_name, credits.

SELECT course_id,
course_name,
credits

FROM courses_table

ORDER BY credits DESC

LIMIT 3;

results_table

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

enrollments-table			
enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

09. Get the maximum, minimum, and average grade across all enrollments.

EXPECTED COLUMNS: max_grade, min_grade, avg_grade

```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments-table;
```

results-table		
max_grade	min_grade	avg_grade
90	78	84.4

10. Count how many enrollments exist per course.
EXPECTED COLUMNS: course_id, enrollment_count

```
SELECT course_id,
       COUNT(enrollment_id) AS enrollment_count
FROM enrollments-table
GROUP BY course_id;
```

results-table	
course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Salaries_table

employee_id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Terry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11. Find total salary and total bonus per department
 EXPECTED COLUMNS: department, total_salary, total_bonus

SELECT department,
 SUM(salary) AS total_salary,
 SUM(bonus) AS total_bonus

FROM salaries_table
 GROUP BY department;

result_table

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70 000	6000

12. Show departments where average salary is above 55000
 EXPECTED COLUMNS: department, avg_salary.

SELECT department,
 AVG(salary) AS avg_salary
 FROM salaries_table
 GROUP BY department
 HAVING AVG(salary) > 55000;

results_table

department	avg_salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 60000
 EXPECTED COLUMNS: employee_id, name, salary, bonus, total_compensation

```

SELECT employee_id,
       name,
       salary,
       bonus,
       salary + bonus AS total_compensation
FROM salaries_table
WHERE salary + bonus > 60000;
  
```

results_table

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5300	67500

projects_table

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR portal	HR	50000

14. Show total and average budget per department. Only include departments with average budget above 70000
 EXPECTED COLUMNS: department, total_budget, avg_budget


```

SELECT department,
       SUM (budget) AS Total_budget,
       AVG (budget) AS avg_budget
FROM projects_table
GROUP BY department
HAVING AVG (budget) > 70 000;

```

results_table

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50 000 and 120 000, excluding the Marketing department.
 EXPECTED COLUMNS: project_id, project_name, department, budget.

```

SELECT *
FROM projects_table
WHERE NOT department = 'Marketing'
AND budget BETWEEN 50 000 AND 120 000;

```

results_table

1	AI App	IT	120 000
2	Payroll system	Finance	80 000
5	HR portal	HR	50 000